44/50 88%

ABERDARE TECHNICAL COLLEGE
DEPARTMENT: INFORMATION COMMUNICATION TECHNOLOGY (ICT)
UNIT NAME: PERFORM PROGRAMMING PRINCIPLES
UNIT CODE: 0613 451 05A
ASSESSMENT: CONTINUOUS ASSESSMENT TEST (CAT 2)
TIME: 1 HOUR 30 MINUTES
TOTAL MARKS: 50

INSTRUCTIONS TO THE CANDIDATE MARCUS 828

i. Answer ALL questions.
ii. The marks for each question are shown in brackets.
iii. Write all programs neatly in C and include comments where necessary.
iv. Mobile phones and notes are not allowed in the examination room.

QUESTIONS

1. Define the term *control structure* and name the three main types of control structures in C. (3 marks)

2. Name three input functions in C and state what each function reads. (3 marks)

3. State four criteria used when selecting a data type based on system requirements. (4 marks)

4. Differentiate between a function declaration (prototype) and a function definition. (3 marks)

5. Study the array declaration below and answer the questions that follow. (3 marks)

int marks[5] = {50, 60, 70, 80, 90};

(a) State the values stored in marks[2] and marks[4]. (2 marks)

(b) State the first and last valid indexes of the array. (1 mark)

6. A three-dimensional array is declared as shown below. (4 marks)

int a[2][3][4];

(a) Explain the meaning of each of the three indexes. (3 marks)

(b) Calculate the total number of elements the array can store. (1 mark)

7. Study the two-dimensional array below and answer the questions that follow. (5 marks)

```
int scores[3][4] = {
    {12, 8, 15, 20},
    {17, 18, 9, 11},
    {14, 6, 19, 5}
};
```

(a) State the number of rows and columns in the array. (2 marks) 3 rows · 4 columns

(b) Write the values stored in scores[1][1] and scores[2][3]. (2 marks) 18, 5

(c) Write a C statement that changes the element in row 2, column 0 to 30. (1 mark)

8. Study the two programs below and answer the questions that follow. (5 marks)

Program 1

```
#include <stdio.h>

void change(int x)
{
   x = 50;
}

int main()
{
   int number = 10;
   change(number);
   printf("Number = %d", number);
   return 0;
}
```

Program 2

```
#include <stdio.h>

void change(int *x)
{
   *x = 50;
}

int main()
{
   int number = 10;
   change(&number);
   printf("Number = %d", number);
   return 0;
}
```

(a) State the output of Program 1 and Program 2. (2 marks)

(b) Name the method of passing arguments used in each program. (2 marks)

(c) Explain why the outputs of the two programs are different. (1 mark)

9. Select the most suitable C data type for storing each of the following data and give one reason to justify each choice. (5 marks)

(a) A student's age.

(b) The population of a country exceeding 2.1 billion.

(c) A percentage score from 0 to 100 on a device with very limited memory.

(d) A measurement requiring approximately 15 digits of precision in a scientific calculation.

(e) A single grade letter such as 'A'.

10. Write a complete C program that accepts a student's marks from the keyboard and uses an if-else statement to display the grade according to the following criteria (3 marks)

- 80 and above : Grade A
- 70 to 79 : Grade B
- 50 to 69 : Grade C
- Below 50 : Fail

11. Write a complete C program that performs the following tasks (6 marks)

(a) Defines a user defined function called multiply that accepts two integers and returns their product. (3 marks)

(b) Includes a function prototype for the function. (1 mark)

(c) In the main function, accepts two integers from the user using scanf(), calls the multiply() function, and displays the result. (6 marks)

1. A control structure is a Programming construct that determines the order in which statements in a program are executed.
   i. Sequential ✓
   ii. Selection ✓
   iii. Iteration ✓ 3/3

2. Scanf() reads formatted input from the standard input.
   (getchar()) - reads a single character from the standard input. 2/3
   (gets()) - Read a string from standard input

3. Range of values the variable needs to store.
   The accuracy required ✓
   Storage size available ✓
   Nature of the data. numbers or letters and operation to be performed. 2/4

4. Function declaration tells the compiler the function's name, return type & parameters while function definition contains the complete functions include the body. 2/2

5. a) 3 rows and 4 columns
   b) ~~[1][1] = 10~~, [2][3] = 5

6. a) [2] = 70 [4] = 90 2/2
   b) 0 - first index, 1 - last index 0/1

6. a). [2] = number of layers.
   [3] = number of rows ✓ 2/2
   [4] = number of columns ✓ 1/1
   b) (2)x(3)x(4) = 24 elements

7. a) 3 rows, 4 columns 1/1
   b) [1][1] = 18, [2][3] = 5 2/2
   c) int [2][6] = 30; 1/1

24/24

8 a) Program 1 = 10
Program 2 = 50

b) Program 1 = Call by value
Program 2 = Call by reference

c) In Program 1 changing x inside the function only affects the copy
So the original number remains 10
In Program 2 the function modifies the actual variable through the pointer
and changes reflect in main()

9. a) Student's age: int, the ages white number with only small range of numbers
b) Population of country exceeding 2.1 billion. VARCHAR Population exceeds maximum
value of normal int so needs much larger space.
c) % Score on limited storage. CHAR. The range 0-100 fit in single byte
d) Measurement of approximate 15 digits. Long double. This provides high
needed for such calculation.
e) Single grade letters as 'A'. CHAR. Single character is best stored by Char

10.
```
#include <stdio.h>
int main() {
int marks;
Printf("Enter the students marks:");
Scanf("%d", &marks);
if (marks >= 80) {
Printf("Grade A\n");
}
else if (marks >= 60) {
Printf("Grade B\n");
}
else if (marks >= 60) {
Printf("grade C\n");
}
else {
Printf("fail\n"); }
```

```
#include <stdio.h>
int multiply (int a, int b);
int multiply (int a, int b)
{
return a*b;
}
int main ()  {
int num1, num2, result;
printf ("Enter an integer: ");
scanf ("%d", &num1);
printf ("enter an integer: ");
scanf ("%d", &num2);
result = multiply (num1, num2);
printf ("The Product is: %d\n", result);
return 0;
}
```

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